

Rahul Bouri

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Summary

Machine Learning and AI Research Engineer experienced in production AI systems, scalable ML infrastructure, and multimodal ML, with hands-on work in robotic simulation, VLA policies, and synthetic data generation. Seeking ML/Research Engineering roles focused on scalable ML systems, data pipelines, and robot learning.

Education

Carnegie Mellon University

Master of Science in Robotic Systems Development

Pittsburgh, PA

Aug 2026 – May 2028

Coursework: Robot Learning, Planning and Decision Making, Robot Autonomy, Systems Engineering

Birla Institute of Technology and Science (BITS), Pilani

Pilani, India

Bachelor of Engineering in Mechanical Engineering

Aug 2020 – May 2024

Major Coursework: Linear Algebra, Multivariate Calculus, Control Systems, Mechanisms and Machines

Minor Coursework: Machine Learning, Deep Learning, Applied Statistical Methods, Data Mining

Skills

Programming, ML & Data Science: Python, C++, SQL, PySpark, PyTorch, MLflow, Weights & Biases, Git

Robotics & Infrastructure: ROS2 (MoveIt2, Nav2, SLAMtoolbox, ROS2 Control), MuJoCo, Isaac Sim/Lab, OpenAI Gym, Kubernetes, Docker, AWS/GCP, Linux, MATLAB, CAD

Work Experience

AI Research Engineer - AlphaSense, Bangalore, India

December 2025–July 2026

- Built and deployed a document-grounding service for LLM responses using hybrid lexical and LLM-based extraction with confidence-aware inference; containerized the FastAPI service with Docker and deployed on AWS, achieving **p90 latency <3s at 100 RPS**.
- Developed a centralized registry and standardized interface for **100+ agent workflows**, enabling capability discovery, onboarding, and lifecycle management across workflows handling **30% of production traffic**.

Machine Learning Engineer - OLA Cabs, Bangalore, India

August 2024–November 2025

- Built a natural-language enterprise analytics system using semantic retrieval, SQL generation with validation, and structured error recovery, achieving **92% task success** in human-verified evaluations.
- Productionized an ETA prediction pipeline with real-time feature generation and model drift detection supporting **100K+ daily orders**, achieving **95% of deliveries within ± 2 minutes of the predicted ETA** and reducing cancellations from **26% to 14%**.
- Developed an image segmentation and OCR pipeline for **15+ government-issued documents**, processing **10K+ daily requests at 98% accuracy** – reducing manual verification costs by **INR 5L+/month**.

Robotics Intern - TCS Research, New Delhi, India

June 2023–December 2023

- Implemented RGB-D and LiDAR-based SLAM in **ROS2** for autonomous indoor localization and navigation of an assistive robot.
- Built a multimodal emotion recognition pipeline in **PyTorch**, achieving **74% weighted accuracy**, and integrated predicted emotions into a speech synthesis module for adaptive human-robot interaction.

Projects & Publications

Self-Designed Bipedal Walker – Independent

Jul 2026–Current

- Designing and assembling a **custom 784 g Jetson-class biped** — actuator selection, structural layout, and wiring.
- Building a **real2sim $\rightarrow$ sim2real pipeline** to walk, balance, and reject external pushes on hardware.

Proprioceptive Contact Estimation – Independent

May 2026–Jun 2026

- Estimated external contact torque from **joint encoders alone** — no force/torque sensor or tactile skin.
- EKF turns rigid-body model error into a contact signal for **collision detection and safe interaction**.
(raoulbouri.github.io/projects/proprioceptive-contact-detection)

Agentic LLMs for Question Answering over Tabular Data

Independent · SemEval-2025 Workshop (Task 8: DataBench) @ ACL 2025

Nov 2024–Jan 2025

- Multi-stage NL-to-SQL agent with query verification and iterative refinement; achieved **70.5% on DataBench QA** against a 26% baseline. (aclanthology.org/2025.semeval-1.292)

Improving Narrative Classification and Explanation via Fine Tuned Language Models

Independent · SemEval-2025 Workshop (Task 10) @ ACL 2025

Nov 2024–Jan 2025

- Recall-oriented **BERT fine-tuning refined by GPT-4o**, with ReAct retrieval prompting and taxonomy grounding to curb hallucinated justifications. (aclanthology.org/2025.semeval-1.290)